

Density-dependent changes of the pore properties of the P2X₂ receptor channel

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Ligand-gated ion channels underlie and play important roles in synaptic transmission, and it is generally accepted that the ion channel pores have a rigid structure that enables strict regulation of ion permeation. One exception is the P2X ATP-gated channel. After application of ATP, the ion selectivity of the P2X₂ channel time-dependently changes, i.e. permeability to large cations gradually increases, and there is significant cell-to-cell variation in the intensity of inward rectification. Here we show P2X₂ channel properties are correlated with the expression level: increasing P2X₂ expression level in oocytes increases permeability to large cations, decreases inward rectification and increases ligand sensitivity. We also observed that the inward rectification changed in a dose-dependent manner, i.e. when low concentration of ATP was applied to an oocyte with a high expression level, the intensity of inward rectification of the evoked current was weak. Taken together, these results show that the pore properties of P2X₂ channel are not static but change dynamically depending on the open channel density. Furthermore, we identified by mutagenesis study that Ile328 located at the outer mouth of the pore is critical for the density-dependent changes of P2X₂. Our findings suggest synaptic transmission can be modulated by the local density-dependent changes of channel properties caused, for example, by the presence of clustering molecules.

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P2X receptors are ATP-gated ion channels that are widely distributed in the peripheral and central nervous system. They are non-selective cation channels (Virginio *et al.* 1999) with the property of inward rectification (Khakh *et al.* 1995; Evans *et al.* 1996; Zhou & Hume, 1998). It is known that they play critical roles in fast synaptic transmission (Edwards *et al.* 1992; Evans *et al.* 1992) and in presynaptic modulation (Gu & MacDermott, 1997; Khakh & Henderson, 1998; Kato & Shigetomi, 2001).

The primary structure of P2X receptors was determined by the isolation of the first cDNA by the expression cloning method (Brake *et al.* 1994; Valera *et al.* 1994), and seven types of P2X cDNAs have been cloned so far (North & Barnard, 1997; North, 2002). All of them have two transmembrane regions with a large extracellular loop, and the functional unit is reported to be composed of three subunits based on the analysis of the weight of the non-denatured proteins (Nicke *et al.* 1998). This structure,

a trimer of two transmembrane type subunits, is in clear contrast with other ligand-gated channels such as nicotinic acetylcholine (nACh) receptors or glutamate receptors (Khakh, 2001), suggesting the possibility of functional differences.

The P2X receptor channel is unique not only from a structural but also from a functional point of view. First, its ion selectivity changes dynamically within a time frame measured in seconds (Surprenant *et al.* 1996; Khakh & Lester, 1999; Virginio *et al.* 1999; Khakh *et al.* 2001; Eickhorst *et al.* 2002), i.e. only small cations such as Na⁺ and K⁺ can pass through the channel pore early after ATP application, but cations as large as *N*-methyl-D-glucamine (NMDG) pass through the channel later (Virginio *et al.* 1999; Eickhorst *et al.* 2002). Second, there is a significant cell-to-cell variation in the intensity of the inward rectification through the P2X₂ channel (Khakh *et al.* 1995; Evans *et al.* 1996). Third, the channels interact with nACh receptors directly on the membrane, resulting

in inhibition of the nACh receptor current (Nakazawa, 1994; Khakh *et al.* 2000).

As it is generally accepted that pore properties of ion channels such as ion selectivity and rectification are determined strictly by the tight molecular base like those of KcsA (Doyle *et al.* 1998; Zhou *et al.* 2001) or Kir (Nichols & Lopatin, 1997; Kubo & Murata, 2001; Fujiwara & Kubo, 2002; Nishida & MacKinnon, 2002), the three unique properties of P2X receptors, which account for the dynamics and flexibility of the pore, are all the more noteworthy.

We initially aimed to identify structural determinants of P2X₂ receptors involved in the property of inward rectification and prepared various mutants. However, reliable determination of the mutant phenotype was hard, because the intensity of inward rectification varied significantly even in the wild-type (WT). We later noticed that the variation of the inward rectification intensity was due to a difference in expression level. Khakh *et al.* (2000) also reported the significance of the expression level for the functional interaction between P2X₂ and nACh receptors. Triggered by these observations, we decided to focus on the correlation of the expression density and the channel properties of the P2X₂ receptor. In the present study, we used *Xenopus* oocytes as an expression system, and analysed channel properties such as the ion selectivity, inward rectification and ATP sensitivity by recording ATP-evoked current under two-electrode voltage clamp from oocytes with various express levels.

Methods

Molecular biology

A BamHI–NotI fragment of the original rat P2X₂ receptor cDNA (Brake *et al.* 1994) was subcloned into pBluescript vector. cRNA was then prepared from the linearized plasmid cDNA using an RNA transcription kit (Stratagene). The single point mutant was made with a QuickChange site-directed mutagenesis kit (Stratagene) and confirmed by sequencing.

Preparation of *Xenopus* oocytes

Xenopus oocytes were collected from frogs anaesthetized in water containing 0.15% tricaine; after the final collection the frogs were killed by decapitation. Isolated oocytes were treated with collagenase (2 mg ml^{−1}, type 1, Sigma), after which oocytes of similar size at stage V were injected with approximately 50 nl of cRNA solution. Serial dilutions (1 : 1 to 1 : 1000) of the cRNA in water enabled induction of various levels of P2X₂ receptor expression. The injected

oocytes were incubated for 1–2 days at 17°C in frog Ringer solution.

All experiments conformed with the guidelines of the Animal care Committees of Tokyo Medical and Dental University and the National Institute for Physiological Sciences.

Electrophysiology

Macroscopic currents were recorded using the two-electrode voltage clamp technique with a bath-clamp amplifier (OC-725C, Warner Instruments, Inc., Hamden, CT, USA). Stimulation, data acquisition and data analysis were all done on a Pentium-based computer using Digidata 1322A and pCLAMP 8.2 software (Axon Instruments, Union City, CA, USA). All recordings were obtained at room temperature (20–23°C). Intracellular glass microelectrodes were filled with 3 M potassium acetate with 10 mM KCl (pH 7.2). The microelectrode resistances ranged from 0.1 to 0.3 MΩ. Two Ag–AgCl pellets (Warner Instruments) were used to pass the bath current and sense the bath voltage. The voltage-sensing electrode was placed near the oocyte (approximately 2 mm away) on the same side as the voltage-recording microelectrode. The bath current-passing pellet and the current injection microelectrode were placed on the other side.

The recording bath solution contained 98 mM NaCl, 1 mM MgCl₂ and 5 mM Hepes (pH 7.35–7.40). For the experiment on the permeability of NMDG and K⁺ in Figs 1 and 2, the bath solution contained 98 mM NMDG, 85 mM HCl, 1 mM MgCl₂ and 5 mM Hepes (pH 7.35–7.40), or 98 mM KCl, 1 mM MgCl₂ and 5 mM Hepes (pH 7.35–7.40). Ca²⁺ was not included in the bath solution to avoid channel desensitization and a variety of secondary intracellular effects. ATP disodium salt (Sigma) was dissolved in bath solution just before each experiment. When applied to cells, one-fifth bath volume of 5 times concentrated solution was pipetted into the bath; complete exchange of solution around the oocyte was confirmed within ~0.5 s.

Current–voltage relationships were recorded at steady states (at ~5 s after ATP application) by applying slow ramp pulses from −75 to +75 mV over 1.5 s (Figs 1D–F, 3A–E, 5A–C, and 6A, B and D), reverse ramp pulses from +75 mV to −75 mV over 1.5 s (Fig. 3F), and ramp pulses from −105 to 75 mV over 1.8 s (Fig. 1A–C). Reversal potentials in Fig. 2 were recorded by applying 360 ms ramp pulses from −105 to 0 mV repeatedly every 500 ms for 30 s after ATP application. The data in Fig. 4 were recorded by applying a set of step pulses at a steady state (at ~5 s after ATP application); cells were held at 0 mV, then stepped to

−80 mV for 150 ms, after which 300-ms test pulses from +60 to −30 mV were applied in 10 mV decrements. All recorded currents, except for those of I328F mutants which showed a basally active component, were leak subtracted, i.e. the background leak current before ATP application was subtracted from the ATP-evoked current. The leak current was monitored again after washing out the ATP, and records showing a significant increase in leak current were discarded. Actual clamped membrane potentials were monitored during the recordings, and data with an error of over 2 mV from the command potentials were discarded (Kubo & Murata, 2001; Fujiwara & Kubo, 2002).

Data analysis

Data were analysed using Clampfit 8 (Axon Instruments, Inc) or Igor Pro (WaveMetrics, Inc., Lake Oswego, OR, USA). Ion permeability ratios in Figs 1C and F and 2C were calculated from the shift of the reversal potential using the function $P_X/P_Y = \exp(\Delta E_{\text{rev}} F/RT)$, where ΔE_{rev} is the difference of the reversal potentials between $E_{\text{rev},X}$ and $E_{\text{rev},Y}$ (Khakh *et al.* 1999). As an index of the intensity of inward rectification in Figs 3C–F, 5C and 6A and D the ratio of the current amplitudes at +40 mV and −60 mV was calculated. Decaying outward currents in Figs 4A and D could be fitted satisfactorily with a single exponential function, and the fitting qualities were not improved by fitting with a double exponential function. K_d and the Hill coefficient in Figs 5 and 6 were calculated from the current amplitudes at a steady state in Na⁺-based external solution at −60 mV obtained by applying various [ATP] by fitting the [ATP]–response relationship with the Hill equation. Correlations of two parameters were calculated by simple linear regression; the regression lines and the correlation coefficients are shown on the graphs. Two regression groups were compared statistically using their regression slopes in Figs 3D and 6A and D. Data from the same batch of oocytes were used for comparison of phenotypes because properties such as the inward rectification differed from batch to batch. Nevertheless, similar tendencies were reproducibly observed in 11 batches of oocytes.

Results

Correlative changes of the ion selectivity with the expression level

We have observed that the extent of the change in ion selectivity correlates with the level of the channel's expression. After establishing a two-electrode voltage clamp configuration, we applied a saturating concentration (100 μM) of ATP to oocytes expressing

low (Fig. 1A) or high (Fig. 1B) levels of P2X₂ in Na⁺-based or NMDG-based external solution. We obtained current–voltage (*I*–*V*) relationships by applying slow (1.8 s) ramp pulses after 8 s of ATP application (Figs 1A and B). In an oocyte with a high expression level the value of the reversal potential in NMDG-based external solution was more depolarized than that in an oocyte with a low expression level, indicative of higher NMDG permeability (Figs 1A and B). $P_{\text{NMDG}}/P_{\text{Na}}$ ratios at 8 s showed a positive correlation with the expression level (Fig. 1C). We performed the same recording in oocytes expressing low (Fig. 1D) or high (Fig. 1E) levels of P2X₂ in Na⁺-based or K⁺-based external solution. P_K/P_{Na} ratios showed no correlation with the expression level (Fig. 1F). We also found that the inward rectification in oocytes expressing a high level of P2X₂ was weaker than that in those expressing a low level (Fig. 1A and B, or D and E), and this is analysed further in the next section.

Next we examined whether or not the time-dependent change of permeability to NMDG was dependent on the expression level. By applying ramp pulses repeatedly, we observed that there was a time-dependent shift in the reversal potential toward depolarization in NMDG-based solution that was much more pronounced in cells expressing high levels of P2X₂ (Fig. 2A). Changes of reversal potentials in Fig. 2A are shown in Fig. 2B. The calculated permeability ratios of NMDG and Na⁺ ($P_{\text{NMDG}}/P_{\text{Na}}$) at 3 s, 10 s and 25 s after ATP application were all positively correlated with the inward current amplitude (Fig. 2C). In short, P2X₂ channels showed low permeability to NMDG immediately after application of ATP even when the expression level was high. The permeability to NMDG increased depending on time and the increase was notable when the expression level was high.

Time-dependent increase in NMDG permeation was reported previously for P2X₇ (Surprenant *et al.* 1996) and P2X₂ (Virginio *et al.* 1999; Eickhorst *et al.* 2002), and it was shown that the changes are not due to ion flux. The novel finding of the present work is that the change is more prominent when the expression level is high. As we used a similar recording condition to the previous studies, we think the time-dependent changes are not due to ion flux.

Correlation between the inward rectification intensity and the expression level

We next investigated the relationships between the inward rectification intensity of P2X₂ and the expression level. Current traces were obtained from oocytes expressing

various levels of P2X₂ by applying ramp pulses while the cells were in a steady state after application of 100 μ M ATP in Na⁺-based external solution (Fig. 3A). The normalized *I*-*V* relationships showed that the intensity of the inward rectification varied from cell to cell (Fig. 3B). When the accumulated data were plotted, the rectification index ($I_{+40 \text{ mV}}/I_{-60 \text{ mV}}$) showed a positive correlation with the inward current amplitude (Fig. 3C). This tendency was also observed in K⁺-based external solution (Fig. 3D). In addition, there were no significant differences between two lines calculated by simple linear regression with the data in Na⁺-based external solution and in K⁺-based external solution.

In these experiments we induced variation in the level of P2X₂ expression by injecting oocytes with different amounts of cRNA. To confirm that the amount of injected cRNA did not influence the intensity of rectification, we analysed the correlation by recording from oocytes injected

with the same amount of cRNA at various times after injection (Fig. 3E). In this experiment, the rectification index again correlated positively with the expression level (Fig. 3E), showing that the variation in rectification was not dependent on the injected amount of cRNA.

We also applied reverse ramp pulses in order to confirm that the intensity of rectification was not dependent on pulse protocols or ion fluxes as shown in Fig. 3F. We confirmed furthermore that ion fluxes could not affect the density-dependent change of inward rectification using rapid ramp pulses (360 ms), which minimize the ion flux, and frequently repeated ramp pulses (10 pulses during 15 s), which maximize it (data not shown). Taking these results together with those using K⁺ as a charge carrier, and reverse or short or repetitive pulses, we concluded that the expression level-dependent changes of inward rectification intensity was not due to the ion flux but due to the expression level itself.

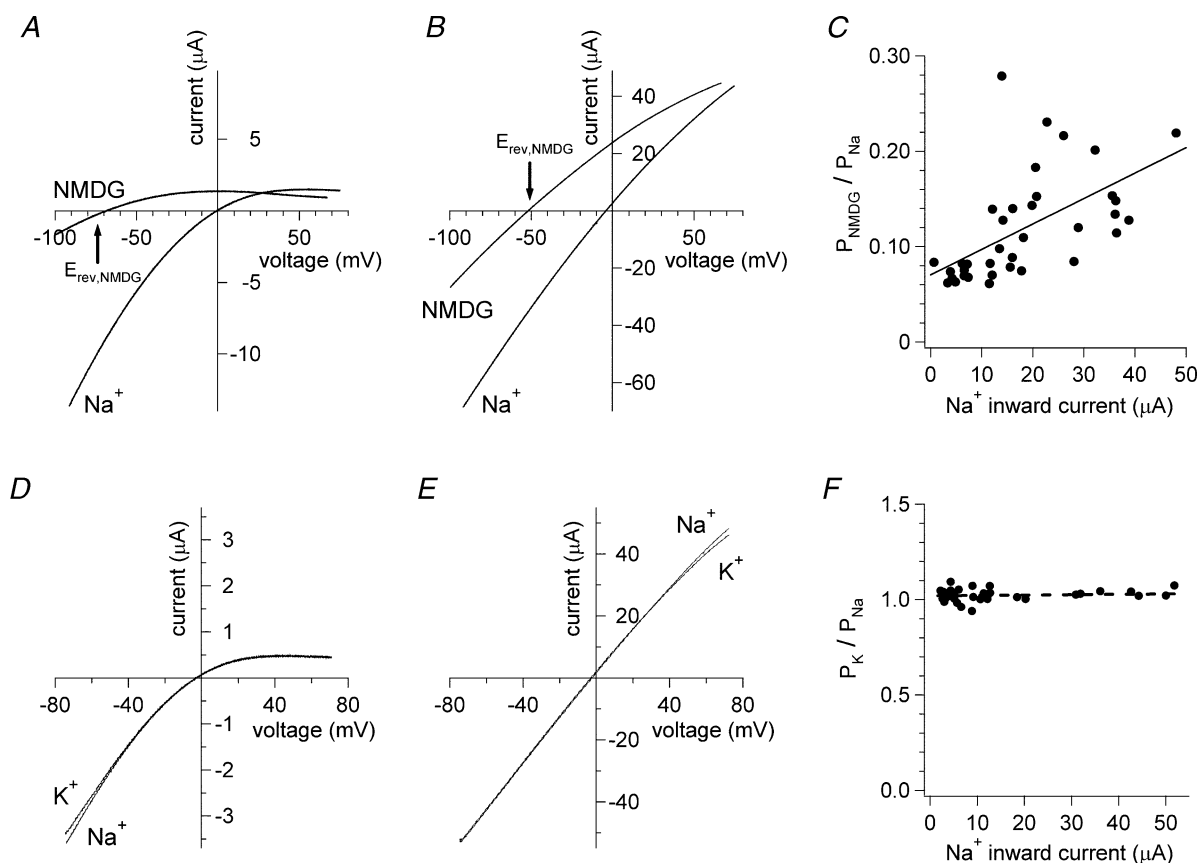


Figure 1. Changes in the ion selectivity of the P2X₂ receptor channel were dependent on the receptor's expression level

A and B, current-voltage relationships obtained 8 s after ATP application by applying ramp pulses to oocytes expressing low (A) or high (B) levels of P2X₂ receptor in Na⁺-based or NMDG-based external solution. C, there was a positive correlation between P_{NMDG}/P_{Na} and P2X₂ receptor expression level. D and E, current-voltage relationships were obtained from oocytes expressing low (D) or high (E) levels of P2X₂ receptor in Na⁺-based or K⁺-based external solution. F, there was no significant correlation between P_K/P_{Na} and P2X₂ receptor expression.

Analysis of the properties of the outward current and their correlation with the expression level

We analysed outward currents evoked by depolarizing step pulses at steady states after application of $100\ \mu\text{M}$ ATP. Once inward currents elicited by application of $100\ \mu\text{M}$ ATP reached a steady level, outward currents evoked by depolarizing step pulses showed a single exponential decay to a steady level. We measured the amplitudes of the inward current (I_{inward} at $-80\ \text{mV}$), the initial level at the beginning of the depolarizing pulses to $+60\ \text{mV}$ (I_{initial}) and the steady level at $+60\ \text{mV}$ (I_{steady}). The time constant of decay (τ at $+60\ \text{mV}$) was calculated by fitting the trace with a single exponential function (Fig. 4A). We then analysed the correlations between these parameters and the inward current amplitude, i.e. the level of P2X₂ expression. $I_{\text{initial}}/I_{\text{inward}}$ (Fig. 4B, filled triangles with a continuous line), $I_{\text{steady}}/I_{\text{inward}}$ (Fig. 4B, open squares with a dotted line) and $I_{\text{steady}}/I_{\text{initial}}$ (Fig. 4C) all became larger as the level of P2X₂ expression was increased. τ remained virtually unchanged and showed no significant correlation with P2X₂ expression (Fig. 4D). Assuming that the current decay reflects the channel's transition from an open to a closed state, we calculated the rate constants between the two states. The rate constant for the transition to the

closed state (α) showed a negative correlation with the level of P2X₂ expression, while that for the transition to the open state (β) showed a positive correlation (Fig. 4E). The constant τ -value reflects that the sum of α and β remains constant. In summary, α and β changed gradually in an opposite manner in accordance with the change in the expression level, keeping the sum constant. A similar tendency was observed at other potentials, ranging from $+10\ \text{mV}$ to $+50\ \text{mV}$ (not shown).

How can these expression level-dependent changes be explained? Here we propose two possible models, the 'gradual change model' and the 'mixed population model'. In the gradual change model, we assume that all channels at a given expression level have the same properties, and that the properties change gradually in a manner corresponding to the expression level. Although this model appears to explain the gradual changes of α and β simply, it suffers from the unnatural assumption that the channel protein has an infinite number of functional states. Also, this model cannot explain well why the sum of α and β remains constant. In the second model, the mixed population model, we assume the coexistence of two different types of channel, one with a decaying outward current and another with no decaying outward current. The observed outward currents reflect the sum of these

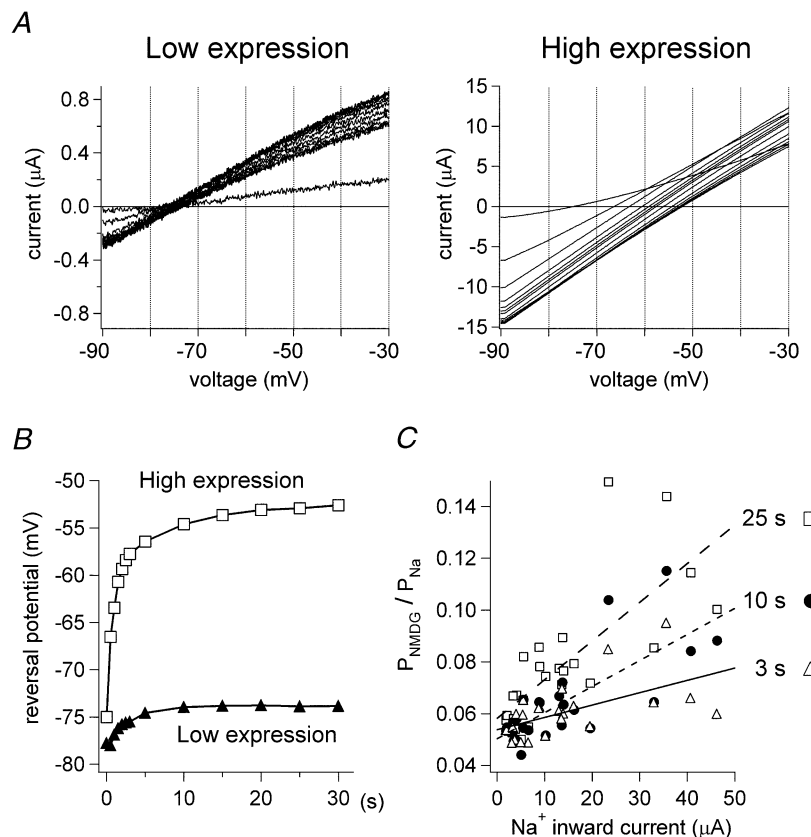


Figure 2. The effect of the P2X₂ receptor's expression level on time-dependent increase in the permeability to NMDG

A, current traces recorded 0.5, 1, 1.5, 2, 2.5, 3, 5, 10, 15, 20, 25 and 30 s after applying ATP to oocytes expressing low (left) and high (right) levels of P2X₂ receptor. B, time-dependent shifts of the reversal potentials were obtained from oocytes expressing low ($\blacktriangle$) and high ($\square$) levels of P2X₂. C, correlation between the expression level of P2X₂ receptor and $P_{\text{NMDG}}/P_{\text{Na}}$ at 3 s (Δ), 10 s ($\bullet$) and 25 s ($\square$) after ATP application. Calculated regression lines are also shown.

two components, and the fraction composed of the non-decaying type increases in accordance with the increases in the level of P2X₂ expression (Fig. 4C). This gradual increase in the fraction of non-decaying type explains the gradual decrease in α , gradual increase in β and constant sum of them. We think this second model is more natural and likely, but we have no conclusive evidence to exclude or to prove one of these two possibilities.

Open channel density dependent dynamic changes of the inward rectification intensity

So far, we have shown that pore properties of P2X₂ change correlatively with the expression level. To test whether pore properties of expressed channels are static or can change dynamically, we compared the I - V relationships

of currents induced by various concentration of ATP ([ATP]) in a single oocyte. Even in an identical oocyte with a high expression level, the intensity of rectification changed by applying various [ATP], i.e. low [ATP] induced a current exhibiting strong inward rectification (Fig. 5A and B). Accumulated data showed that the tendency was applicable at all expression levels (Fig. 5C). Taken together, the rectification intensity is correlated with the density of the 'open' channels, rather than with the density of all the channels expressed, i.e. even in an individual oocyte the pore properties of P2X₂ channels expressed were not stable and could be modified dynamically by applying various [ATP].

These results, however, might be explained also by assuming the presence of two static types of channel: a strong rectifier with high ligand sensitivity and a weak

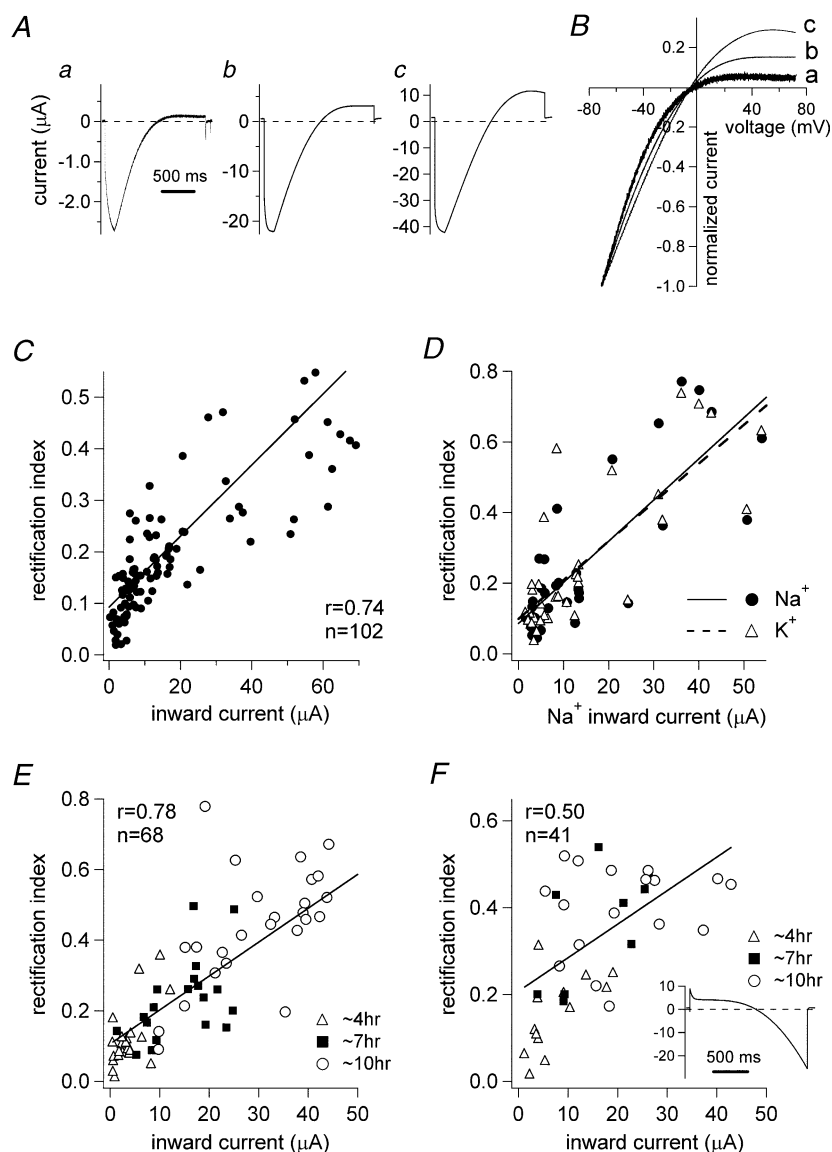


Figure 3. Correlation between the intensity of the inward rectification of P2X₂ channel currents and the level of P2X₂ receptor expression

A, representative currents recorded from oocytes expressing low (a), medium (b) and high (c) levels of P2X₂ receptor. B, normalized I - V relationships. C, correlation between the rectification index ($I_{+40\text{ mV}}/I_{-60\text{ mV}}$) and the current amplitude. D, same as in C, but in this case data obtained in Na⁺-based and K⁺-based external solution were compared. E, same as in C, except that recordings were obtained at various times after injecting the same amount of cRNA into oocytes. F, same as in E, except that reverse ramp pulses from depolarized to hyperpolarized potentials were applied.

rectifier with low ligand sensitivity. In that case, it is not necessary to assume dynamic changes in the open channel properties. To test this possibility we analysed the relationship between ligand sensitivity and the level of P2X₂ expression. Representative concentration–response relationships recorded from four oocytes with various expression levels were shown, and symbols connected by fitted curves are data from the same oocyte (Fig. 5D). The K_d value for ATP, determined from the current amplitudes

measured at a steady-state in Na⁺-based solution, declined as the level of P2X₂ expression increased (Fig. 5E), though the Hill coefficient remained constant at ~2 (Fig. 5F), which is similar to the recently reported findings of Clyne *et al.* (2003). The result that the weak rectifiers present at high expression levels showed a high ligand sensitivity is contrary to the assumption at the beginning of this paragraph. Instead, the results summarized in Fig. 7 can be most naturally explained by assuming that the pore

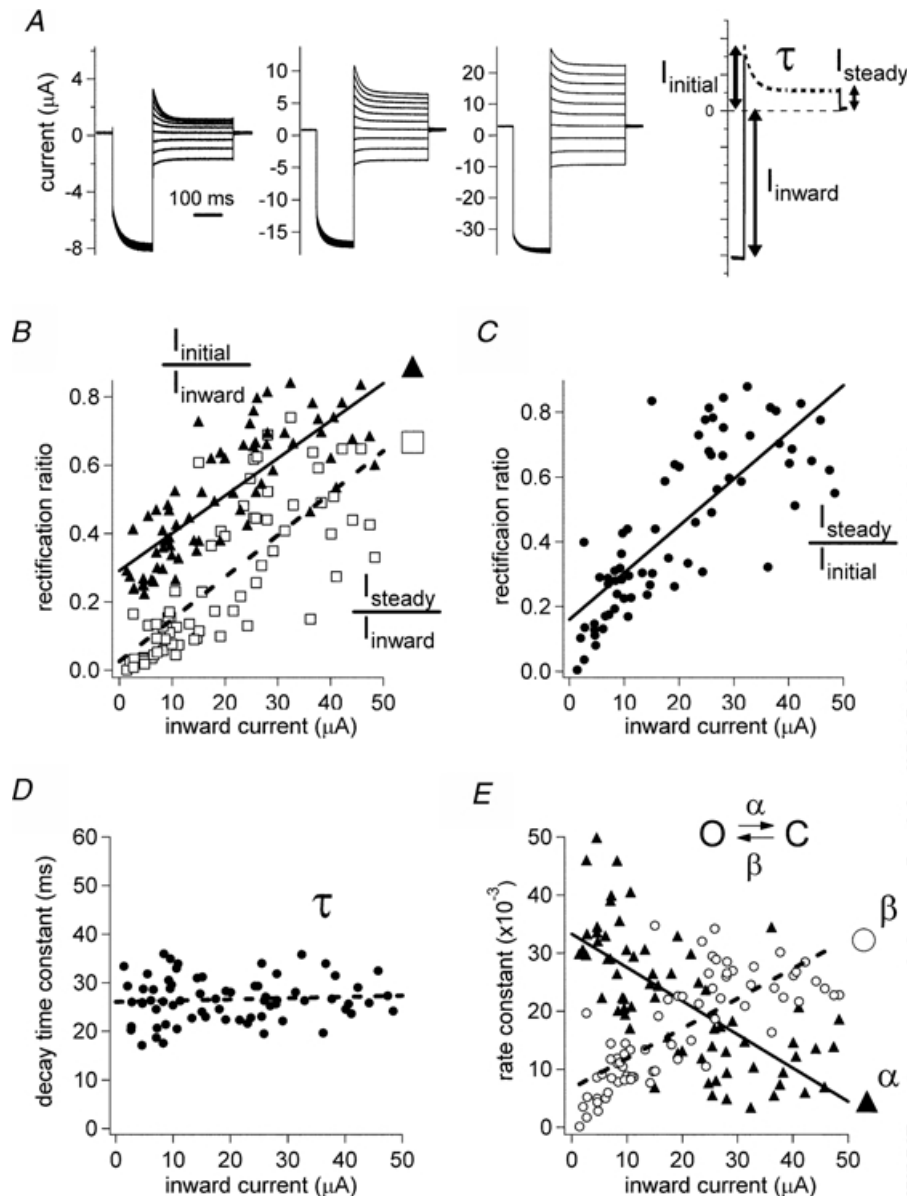


Figure 4. Analysis of the properties of the outward current through the P2X₂ channel

A, representative currents evoked by applying step pulses to oocytes showing increasing (left to right) levels of P2X₂ receptor expression; I_{inward} , I_{initial} , I_{steady} and τ are defined schematically on the right. B, correlation between $I_{\text{initial}}/I_{\text{inward}}$ and I_{inward} and between $I_{\text{steady}}/I_{\text{inward}}$ and I_{inward} . The lines were calculated by simple linear regression (also in C, D and E). C, correlation between $I_{\text{steady}}/I_{\text{initial}}$ and I_{inward} . D, correlation between τ and I_{inward} . E, correlation between calculated α , β and I_{inward} . α and β were calculated from the two equations: $I_{\text{steady}}/I_{\text{initial}} = \beta/(\alpha + \beta)$, $\tau = 1/(\alpha + \beta)$.

properties change dynamically depending on the density of the open channels. Perhaps open channels situated in close proximity interact with one another, leading to a change in the pore properties.

Structural determinant of the density-dependent changes of channel properties

To discover the structural determinant of density dependency, we tried to identify a point mutant whose

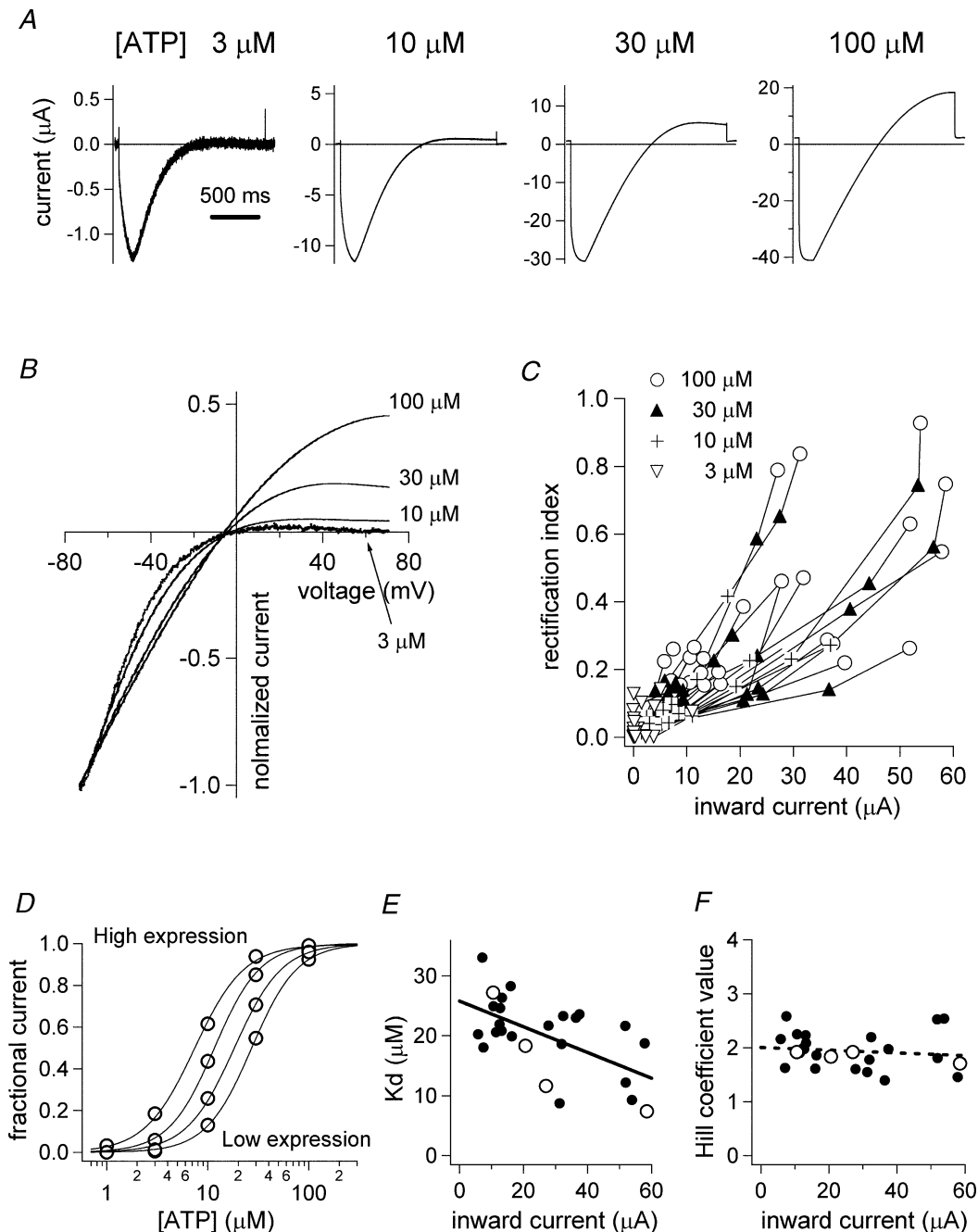


Figure 5. Dependency of the inward rectification intensity on the open channel density

A, representative current recordings evoked by applying ramp pulses to a single oocyte in the presence of various [ATP]. *B*, normalized I - V relationships. *C*, accumulated data showing the correlation of the rectification index with the current amplitude evoked by various [ATP]. Symbols connected by lines are data from the same oocyte. *D*, representative concentration-response relationships fitted with Hill equation. *E*, correlation between K_d and the expression level. The data in *D* were plotted with open circles (also in *F*). *F*, there was no correlation between the Hill coefficient and the expression level.

rectification property does not clearly change depending on the channel density. As the change of the rectification property is expected to involve conformational changes of the pore, we introduced single point mutations systematically to the pore region, and identified I328 as a critical site. I328 is an amino acid residue conserved among all the P2X family and located just above the second transmembrane region; it has been reported by substituted cysteine accessibility analysis to contribute to the permeation pathway and to move during channel opening (Rassendren *et al.* 1997; Egan *et al.* 1998; Jiang *et al.* 2001). The correlation between the current amplitude and the rectification index ($I_{+40 \text{ mV}}/I_{-60 \text{ mV}}$) of the I328C mutant was compared with that of WT (Fig. 6A). I328C showed weaker inward rectification (i.e. a higher rectification index) than WT even in oocytes with a low expression level, and the slope of the regression line of the correlation was significantly less steep than that of WT ($P < 0.001$) (Fig. 6A). In an identical oocyte expressing I328C, the decrease in the rectification index by applying lower [ATP], which was observed in WT, was not remarkable (Fig. 6B). We also analysed the correlation between the K_d value of [ATP] and the expression level (i.e. the inward current amplitude evoked by $100 \mu\text{M}$ ATP)

and observed that the dependency was not apparent in the I328C mutant (Fig. 6C). Taken together, the dependency of both the inward rectification index and the K_d value on the expression level was less clear in I328C than in WT.

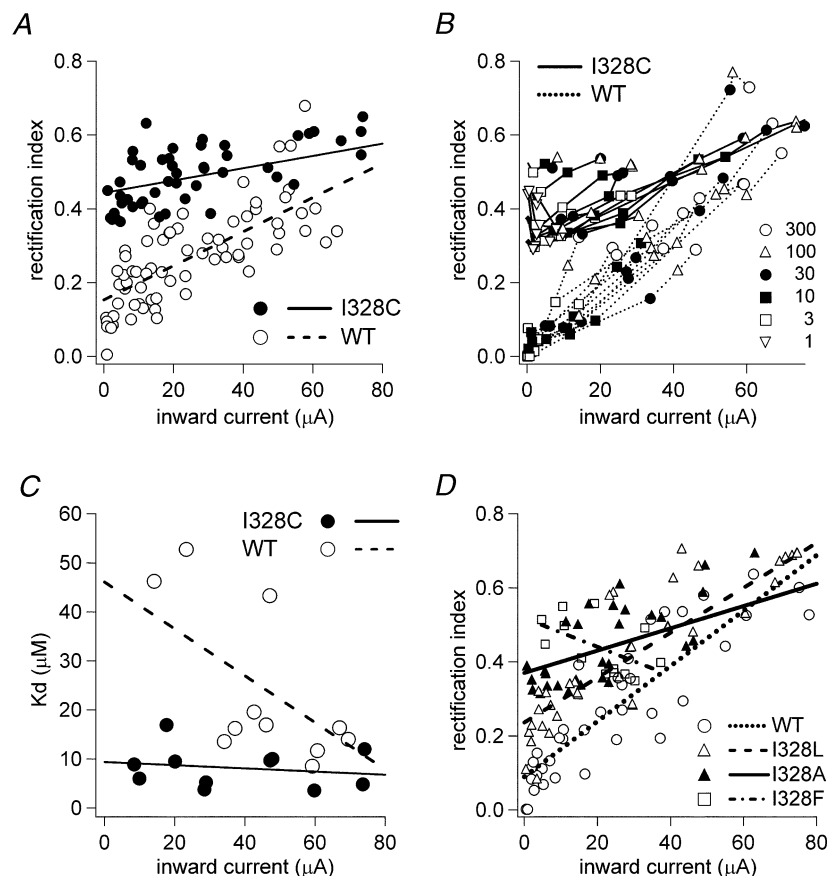
We also made I328A, I328L and I328F mutants, and examined their inward rectification property. The slope of the regression line of I328A ($P < 0.001$), but not of I328L ($P > 0.05$), was significantly less steep than that of WT (Fig. 6D). In the case of the I328F mutant, we could not obtain reliable data for large amplitudes over $40 \mu\text{A}$, because there was a large basally active component on top of the ATP-evoked current. In the obtained data range, the inward rectification intensity of I328F did not show a clear correlation with the current amplitude (Fig. 6D). In summary, the dependency of the inward rectification index on the expression level was affected not only in I328C but also in I328A and possibly in I328F, showing that this residue is critical.

Discussion

In this study we analysed the variation of the properties of P2X₂ receptor channels, focusing especially on the correlation with the expression level. We compared the properties of the current evoked by a saturating dose

Figure 6. Mutants altered the correlation between the inward rectification intensity and the expression density

A, correlation between the rectification index ($I_{+40 \text{ mV}}/I_{-60 \text{ mV}}$) and the current amplitude. WT: ○ and dashed line; I328C mutant: ● and continuous line. **B**, accumulated data showing the correlation of the rectification index with the current amplitude evoked by various [ATP]. Symbols connected by lines are data from the same oocyte. WT: dotted lines; I328C: continuous lines. **C**, correlation between K_d and the expression level. WT: ○ and dashed line; I328C: ● and continuous line. **D**, correlation between the rectification index ($I_{+40 \text{ mV}}/I_{-60 \text{ mV}}$) and the expression level. WT: ○; I328A: ▲; I328L: △ and I328F: □. Calculated regression lines are also shown.



of ATP in oocytes with various expression levels, and observed that there is a strong correlation between the channel properties and the expression level, i.e. in oocytes with higher expression level, the inward rectification intensity of the evoked current was weaker, and the permeability of NMDG was higher.

Dynamic changes of channel properties by mutual interaction between open channels

When we applied low [ATP] to oocytes with a high expression level, the inward rectification of the evoked current was strong, suggesting that the channel properties are not stable but can change dynamically in an open channel density-dependent manner. Virginio *et al.* (1999) have also reported a related finding that the pore dilatation of the P2X₂ receptor could occur more easily by applying higher [ATP]. The schematic drawing shown in Fig. 7 summarizes the dynamic changes in P2X₂ pore properties correlated with expression density and [ATP]. This scheme is based on the aforementioned 'mixed population model', and the explanation of the scheme is as follows. Open P2X₂ receptor channels exist as a mixture of two pore types, O₁ and O₂. Isolated open pores at low expression density or

those activated by applying a low [ATP] are in the O₁ state, which shows strong rectification and low permeability to NMDG. The distances between open channels are reduced by increasing the expression level or by applying a higher [ATP]. Interactions between the open channels lead to a shift to the O₂ state, which shows weak rectification. Depending on the time after ATP application, the channels then shift further to the O₃ state, enabling permeation of NMDG. In contrast to the weak inward rectification property, the high NMDG permeability was achieved in a time-dependent manner after ATP application at high expression level. Therefore we think changes in the rectification property and the NMDG permeability are separable rather than go hand-in-hand.

The key point of this scheme is that interactions between open P2X₂ channels induce changes of pore properties. There are some preceding reports which go with this speculation. (1) Ding & Sachs (2002) analysed the single channel properties of the P2X₂ receptor in membrane patches containing one, two or three channels, and observed that the mean open time of multiple channels is much longer than that of an isolated channel; moreover, the open channel noise is lower in multiple channel patches than in patches with an isolated channel. This means

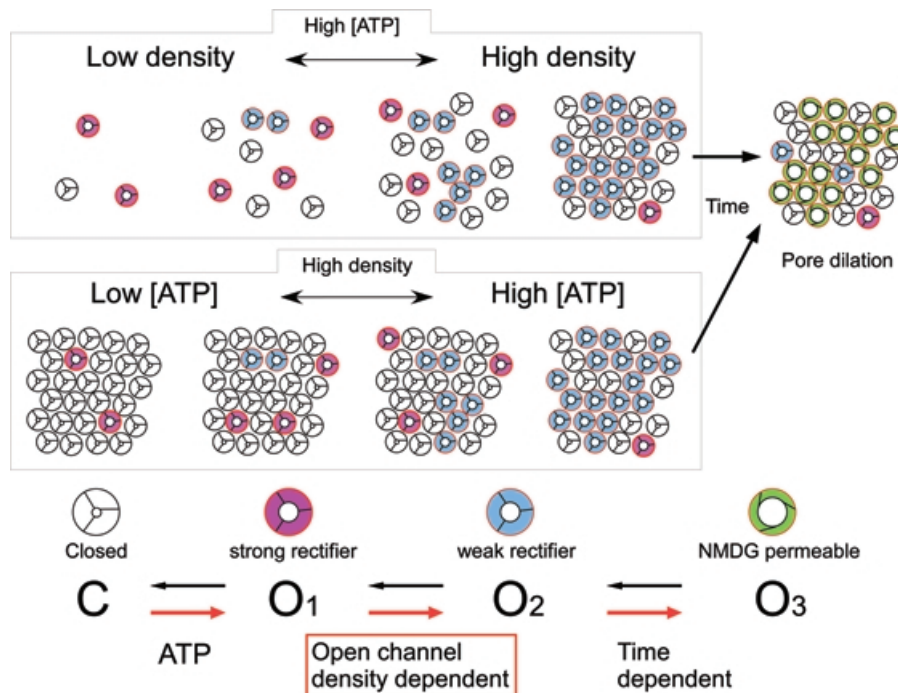


Figure 7. Schematic drawings explaining the dynamic variation of inward rectification intensity depending on the open channel density

The equilibrium is inclined toward the O₂ state, depending on the open channel density. O₁: a state showing a strong inward rectification and a low permeability to NMDG. O₂: a state showing no inward rectification. O₃: a state showing a high permeability to NMDG.

that the single channel properties differ depending on whether it is alone or in a group. (2) It is known that P2X₂ channels inhibit the activities of other ligand-gated channels. When P2X₂ receptors were coexpressed with other ionotropic receptors such as the nACh (Nakazawa, 1994; Khakh *et al.* 2000), GABA_A (Sokolova *et al.* 2001) or 5HT₃ (Barajas-Lopez *et al.* 2002) receptor, and were coactivated by applying both ligands, the amplitudes of the evoked current were smaller than the sum of the current amplitudes when the channels were activated individually. This was especially true when the expression level P2X₂ receptor was high (Khakh *et al.* 2000). These reports are compatible with our hypothesis that adjacent open P2X₂ channels are not independent of one another.

Mechanistic insights for the density-dependent changes

It is noteworthy that the model we proposed in Fig. 7 has a similarity with the alamethicin model. Alamethicin is a channel pore-forming antibiotic, known to change its conductance levels with the increase in the concentration of channel monomer. This phenomenon is reported to be caused by the increase in the chance of collision and by the ensuing changes of the number of multimeric units (Sansom, 1993; Tieleman *et al.* 2002). These reports suggest a general mechanism of homophilic interaction between channels on the membrane, which could be also applied to the P2X₂ receptor channel.

It is of interest to estimate the density or the distance between P2X₂ channels at 'high' expression level, to determine if intermolecular interaction on the membrane could actually occur. With an assumption that the distribution on the membrane is uniform, we roughly estimated the density on the membrane surface. Using a single channel conductance, an open probability (Ding & Sachs, 1999) and an oocyte diameter (~ 1.1 mm), we obtained a value of as low as ~ 25 channels μm^{-2} on the membrane of oocytes with a high expression level, which is apparently lower than in the image drawn in Fig. 7. Since the membrane surface was shown not to be full of P2X₂ channel proteins, direct interaction does not seem to occur. However, if the channels do not sit still but float around on the membrane, there could be a good chance for the accidental collision and interaction of the molecules to occur at relatively high expression density. Also, if the channels are not expressed evenly, but clustered at hot spots with a very high local density, the intermolecular interaction could occur there. This possibility includes an accumulation of receptors by scaffold proteins and/or by lipid rafts. Based on these two reasons, it seems there remains a possibility that a direct interaction could occur.

To examine the possibility that P2X₂ receptors are clustered on the raft and that functional modulation by interaction occur there, we examined the effects of fumonisin B or β -cyclodextrin, which are known to destroy rafts by affecting sphingolipids or cholesterol, respectively. We also examined the effects of nocodazole or colchicine, which are known to disrupt microtubules, assuming that the cytoskeleton plays a critical role in the channel clustering. However, all these drugs did not influence the density-dependent changes of the inward rectification intensity of P2X₂ receptors (data not shown). Therefore, we could not obtain evidences that the high local density and interaction at the raft is critical in *Xenopus* oocytes.

In summary, the expression density on the membrane is not as high as shown in Fig. 7 on the assumption that the distribution is uniform. However, if we assume a diffusion of the receptor on the membrane and a presence of hot spots, there seems to be a good chance for direct interaction. As there is no strong evidence for or against this possibility, this point remains to be elucidated considering other possibilities also.

If P2X₂ 'ionotropic' receptor activation links to a 'metabotropic' pathway, it might be the mechanism of activated receptor density-dependent changes of channel properties. However, we think this possibility is very unlikely as there have been no preceding reports which showed metabotropic coupling of P2X₂ receptor activation, although it has been reported for the AMPA-type glutamate receptor as an exceptional case (Wang *et al.* 1997; Hayashi *et al.* 1999).

Structural determinant for the density-dependent changes

What is the structural determinant for the density-dependent changes of the pore properties? We identified I328 as a critical site, which has been reported to be located in the second transmembrane region and to possibly contribute to the aqueous pore by Cys-scanning mutagenesis studies (Rassendren *et al.* 1997; Egan *et al.* 1998; Jiang *et al.* 2001). We observed that mutations of I328 to other amino acid residues significantly altered the slope of the correlation between the inward rectification intensity and the expression level; the rectification intensity of the I328C or I328A mutant was weak and changed only slightly with the increase in the expression level or in the ATP dose. It is noteworthy that mutations of I328 to small amino acid residues such as Ala and Cys induced clear changes. These results suggest that the pore structures of these mutants correspond to that of WT under the high expression density.

Judging from the location of I328 at the extracellular mouth of the pore, we speculate that it is not a site to detect the channel density information but a critical pore-forming or -modifying site influenced by the channel density, which affects channel properties such as inward rectification. Which other domain is then involved in the detection of the expression density? Eickhorst *et al.* (2002) reported that cytoplasmic C-terminal domains of P2X₂ controlled permeation of NMDG. To examine if this C-terminal cytoplasmic chain is important for the density-dependent changes of pore properties, we made a deletion mutant (R371stop) of the cytoplasmic region. The mutation, however, did not affect the correlation between the rectification intensity and the expression level. We speculate that C-terminus cytoplasmic domains of P2X₂ receptors are important for the transition from state O₂ to state O₃, but not from state O₁ to state O₂.

As there are some assumptions in the interpretation of the results of the Cys-scan mutagenesis, there remains a possibility that I328 does not actually face the permeation pathway (Rassendren *et al.* 1997). It is possible I328 mutation caused the observed changes in the present study by secondary effects on the pore structure or by affecting gating. In any case, we showed that mutations at I328 affected critically the expression level-dependent changes of the channel pore properties, although we do not know yet the mechanisms of homophilic interaction from the structural point of view.

Significance in neurotransmission

It is needless to say that receptor properties such as rectification and ion permeability affect synaptic transmission significantly. For example, it was reported that synaptic transmissions are modulated by the strong inward rectification through neuronal nACh receptors (Sargent, 1993) or by the Ca²⁺ permeation through NMDA-type glutamate receptors (Malenka & Nicoll, 1999). We therefore think that the open channel density-dependent changes of channel properties could serve as a very important modulation mechanism of synaptic function. The presence of various scaffold proteins could elevate the local density of receptors to a very high level, and the turnover of the scaffold proteins could induce fluctuation of the channel density. The density-dependent change of channel properties is not specific to P2X₂ receptors but also observed for voltage-gated channels. Honore *et al.* (1992) reported the inactivation properties and blocker sensitivity of the Shaker K⁺ channel changed significantly depending on the expression level in *Xenopus* oocytes (Honore *et al.* 1992). Shaker K⁺ channels are also known to

show dynamic changes of ion selectivity during activation (Starkus *et al.* 1997) and C-type inactivation (Zheng & Sigworth, 1997). These reports support the possibility that the density-dependent modulation we observed for the P2X₂ receptor can apply in general to receptors and channels at synaptic sites. Taken together, we think there is a good possibility the modulation of synaptic function could actually occur by open channel density-dependent changes of the properties of receptors and channels.

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